

Model-Independent Pricing of Asian Options and the Characterisation of 3-Complete Mixability of Probability Distributions: A Numerical Approach via Optimal Transport

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Abstract

This paper explores an optimal transport based approach to two different problems in probability/mathematical finance. The first concerns martingale optimal transport with an Asian option cost function. We find the support of optimal couplings, varying the strike and a regularisation parameter. The second concerns 3-complete mixability (3-CM) (Rüschendorf–Uckelmann [1], Wang–Wang [2]). We find 3-CM thresholds for symmetric bimodal Gaussian mixtures, symmetric Beta(α, α) laws, and some symmetric point mass laws.

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1 Model-Independent Pricing of Asian Options via Martingale Optimal Transport

1.1 Problem Setup and Approach

Let μ, ν be probability measures on a finite grid $G \subset \mathbb{R}$ with equal means, representing the marginal laws of an underlying asset at two dates $t_1 < t_2$. Write

$$\Pi_M(\mu, \nu) = \left\{ \pi \in \Pi(\mu, \nu) : \int (y - x) \pi(x, dy) = 0 \text{ } \mu\text{-a.s.} \right\}$$

for the set of *martingale couplings*, where $\Pi(\mu, \nu)$ is the set of couplings with marginals μ, ν . For a strike K and regularisation term $\varepsilon \in \mathbb{R}$, define the cost

$$c_{K,\varepsilon}(x, y) = |x + y - K| + \varepsilon |x - y|.$$

The upper bound for the this Asian payoff with the regularisation term is given by

$$\sup_{\pi \in \Pi_M(\mu, \nu)} \mathbb{E}_\pi [c_{K,\varepsilon}(X, Y)]. \quad (1)$$

On the grid G , problem 1 is the linear program

$$\begin{aligned} \max_{P \geq 0} \quad & \sum_{i,j} P_{ij} c_{K,\varepsilon}(x_i, y_j) \\ \text{s.t.} \quad & \sum_j P_{ij} = \mu_i \quad \forall i, \quad \sum_i P_{ij} = \nu_j \quad \forall j, \quad \sum_j P_{ij} y_j = x_i \mu_i \quad \forall i. \end{aligned} \quad (2)$$

We solve (2) on grids of size $N = 150$ with $\mu = N(50, 10^2)$, $\nu = N(50, 20^2)$ (a convex-order pair of marginals, $\nu \succeq_{cx} \mu$, Figure 1) and observe the support of the optimal coupling matrix P as K and ε vary.

1.2 Varying Strike with Gaussian Marginals

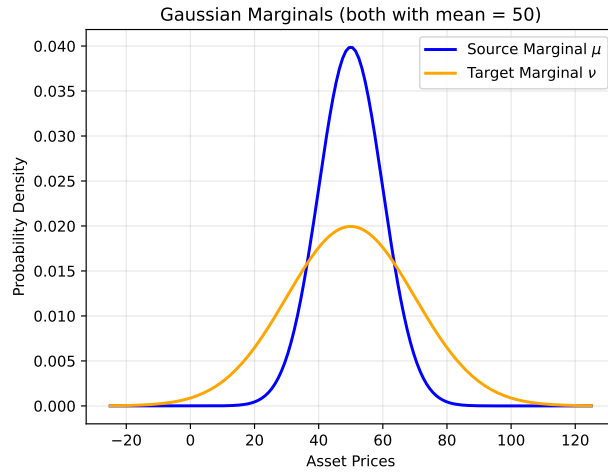


Figure 1: Source marginal $\mu = N(50, 10^2)$ and target marginal $\nu = N(50, 20^2)$; $N = 150$ grid points on $[-25, 125]$.

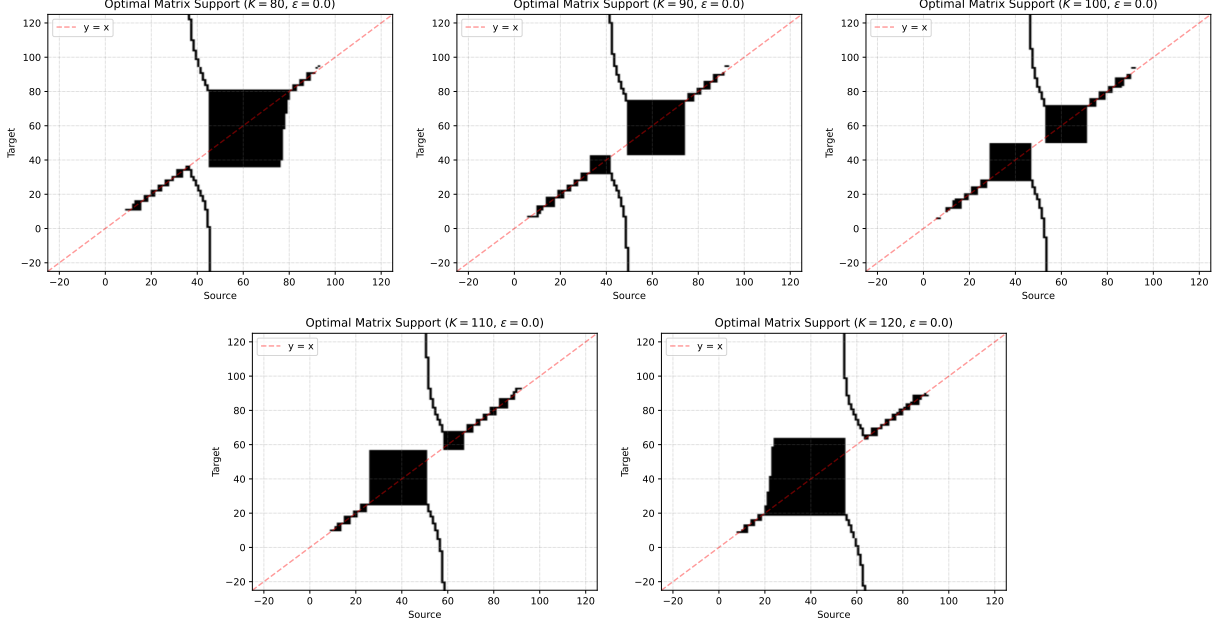


Figure 2: Optimal coupling support for $\varepsilon = 0$ at strikes $K = 80, K = 90, K = 100$ (at-the money), $K = 110$, and $K = 120$. Marginals: $\mu = N(50, 10^2)$, $\nu = N(50, 20^2)$, $N = 150$ grid points on $[-25, 125]$.

Figure 2 shows the optimal coupling matrices across a range of values of K with a fixed $\varepsilon = 0.0$. We observe 2 distinct “cloud” regions on the line $y = x$, with curved arcs extending away from each. As K moves away from the sum of the means of the marginals, the cloud grows on one side and the arcs stretch further along the boundary of the grid. We remark that the extent to which the support stretches along the line $y = x$ at the tails is merely a function of the numerical support threshold ($2\text{e-}6$ in this case).

1.3 Regularisation Parameter Anomaly at $\varepsilon \in \{0, 1\}$

Fixing $K = 100$ and varying $\varepsilon \in [-2, 2]$ over the same Gaussian marginals as in Figure 1, we obtain two different regimes. For most choices of ε the optimal support collapses onto a small number of one-dimensional arcs, while at $\varepsilon = 0$ and $\varepsilon = 1$ the support thickens into two-dimensional “cloud” regions. Figure 3 shows some examples of these one-dimensional arcs at $\varepsilon = -1.5, -0.5, -0.01, 0.01, 0.5, 1.5$.

Considering now the regularisation values $\varepsilon = 0.0$ and $\varepsilon = 1.0$, giving cost functions

$$c_{K,\varepsilon} = c_{100,0}(x, y) = |x + y - 100|$$

and

$$c_{K,\varepsilon} = c_{100,1}(x, y) = |x + y - 100| + |x - y|$$

respectively at $K = 100$, we obtain a “cloud” of strictly positive entries in the optimal transport matrix as in Figure 3. Such a “cloud” formation cannot be observed at any other values of ε . Most notably, one may expect a third such point at $\varepsilon = -1$. Numerically, this is not observed and suggest that the only anomalous points are $\varepsilon = 0, 1$.

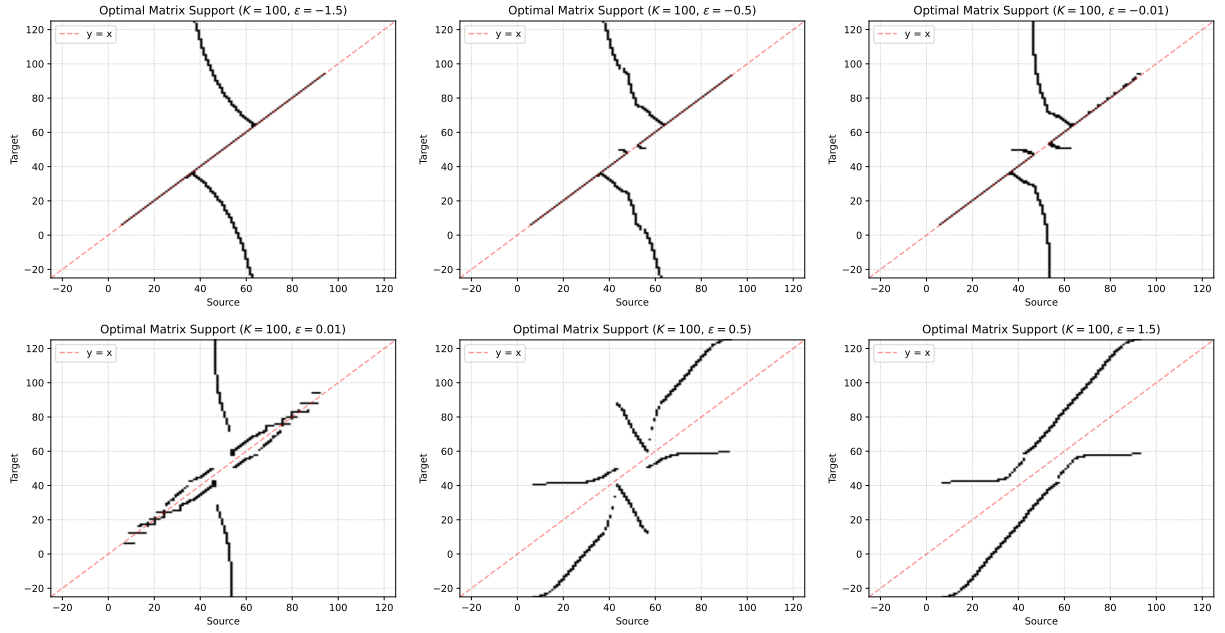


Figure 3: Optimal coupling support at $K = 100$ for $\varepsilon = -1.5, -0.5, -0.01, 0.01, 0.5, 1.5$. Gaussian marginals, each with mean 50 as in Figure 1.

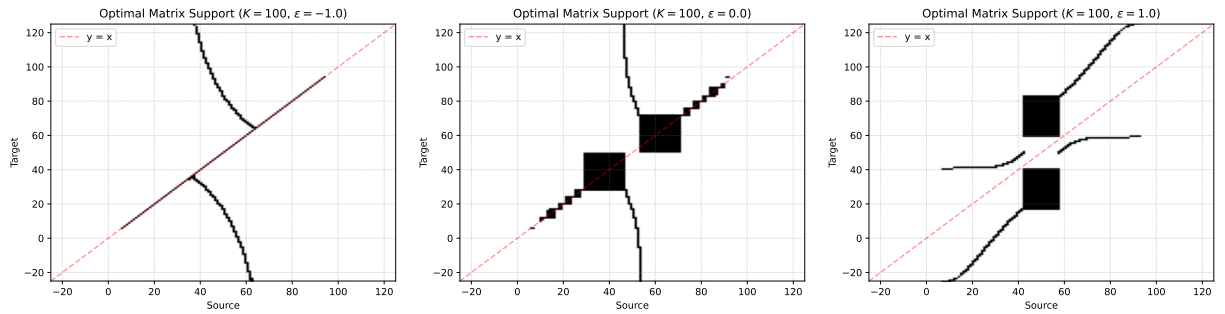


Figure 4: Optimal coupling support at $K = 100$ for $\varepsilon = -1.0, \varepsilon = 0.0$, and $\varepsilon = 1.0$. Same marginals as Figure 2.

1.4 Robustness Under Asymmetric Marginals

We repeat the experiment with asymmetric lognormal marginals as shown in Figure 5, each marginal having mean = 50. Figure 6 shows the support of the optimal coupling with noise filtering threshold at $2e-6$. We note that the general cloud and arc structure remains, although without the symmetry present in the Gaussian case.

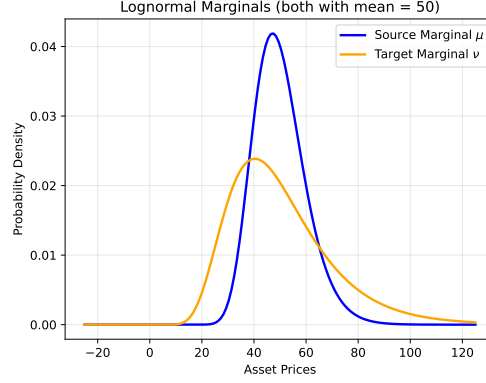


Figure 5: Lognormal marginals (asymmetric) each with mean 50.

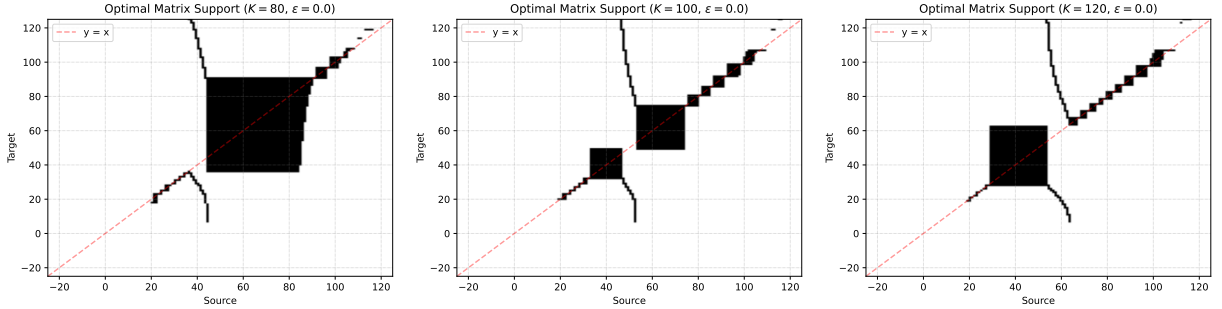


Figure 6: Optimal coupling matrix support in the lognormal marginal case in Figure 5 across strikes $K = 80, 100, 120$.

1.5 The Bimodal Case

We now consider the bimodal case. Consider a unimodal source distribution and a bimodal target distribution as in Figure 7. We consider a skewed tent distribution for the source and a symmetric bimodal tent distribution for the target (both of which have mean = 50). Varying K across values $K = 50, 90, 100, 110, 150$, we obtain the following optimal matrix supports as in Figure 7.

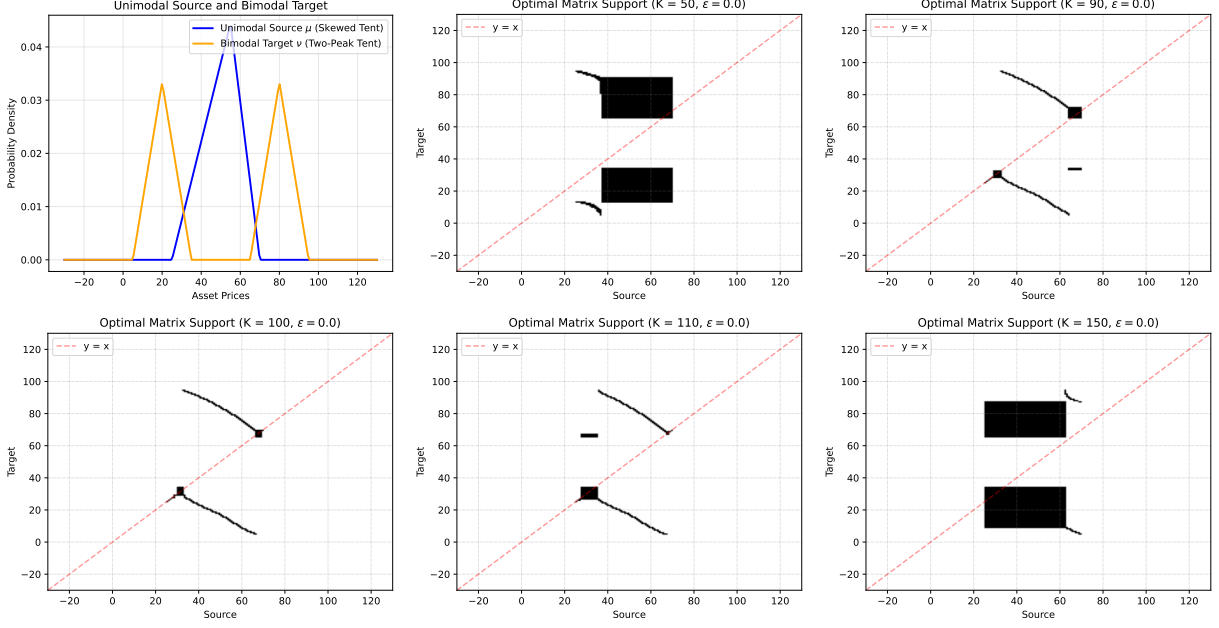


Figure 7: Bimodal tent marginals (top left), optimal coupling matrix supports across strikes $K = 50, 90, 100, 110, 150$

2 Complete Mixability via Optimal Transport

2.1 Problem Setup and Approach

Definition 1 (Complete mixability, [1, 2]). A law μ on $\mathbb{R}$ with finite mean m is *n-completely mixable* (*n*-CM) if there exist $X_1, \dots, X_n \sim \mu$ (not necessarily independent) with

$$X_1 + \dots + X_n = c \quad \text{a.s.}$$

for some $c \in \mathbb{R}$.

Without loss of generality, we restrict to the case $c = 0$. Various sufficient conditions exist for 3-CM, particularly in the case of unimodal marginals (see Rüschendorf–Uckelmann [1] and Wang–Wang [2]). We extend this numerically to cover symmetric bimodal distributions, symmetric beta distributions and symmetric 3- and 5-point mass distributions centred at 0.

In general, we test 3-CM of a discretised law μ on grid G by solving

$$c^*(\mu) = \min_{P \geq 0} \sum_{i,j,k} P_{ijk} |x_i + x_j + x_k - 3\bar{\mu}| \quad \text{s.t. all three marginals of } P \text{ equal } \mu, \quad (3)$$

where $\bar{\mu} = \sum_i x_i \mu_i$, but without loss of generality restrict to the case $\bar{\mu} = 0$. Clearly $c^*(\mu) = 0$ if and only if μ is 3-CM. Resolution and solver parameters are given in Appendix A or stated per case. A threshold of between $10\text{e-}3$ and $10\text{e-}6$ is used to categorise “mixable” or “not mixable” from the numerically computed minimum.

2.2 Sanity check: unimodal symmetric laws

For $\mu \in \{N(0, 3^2), U[-5, 5], \text{Tent}(0, 6)\}$ (all continuous, symmetric, unimodal, hence 3-CM by [1]) solving (3) on grids of size $N = 45$ gives optimal cost $c^* = 1.01 \times 10^{-4}, 1.10 \times 10^{-5}, 2.19 \times 10^{-5}$

respectively, all sufficiently close to the numerical solver noise floor, confirming the numerical method gives the expected result in these known cases. The pairwise optimal tensor projections for each of these marginals is given in Figure 8. Note that since all of these marginals are symmetric, all pairwise projections are identical, we plot all three such projections for each law merely as a sanity check.

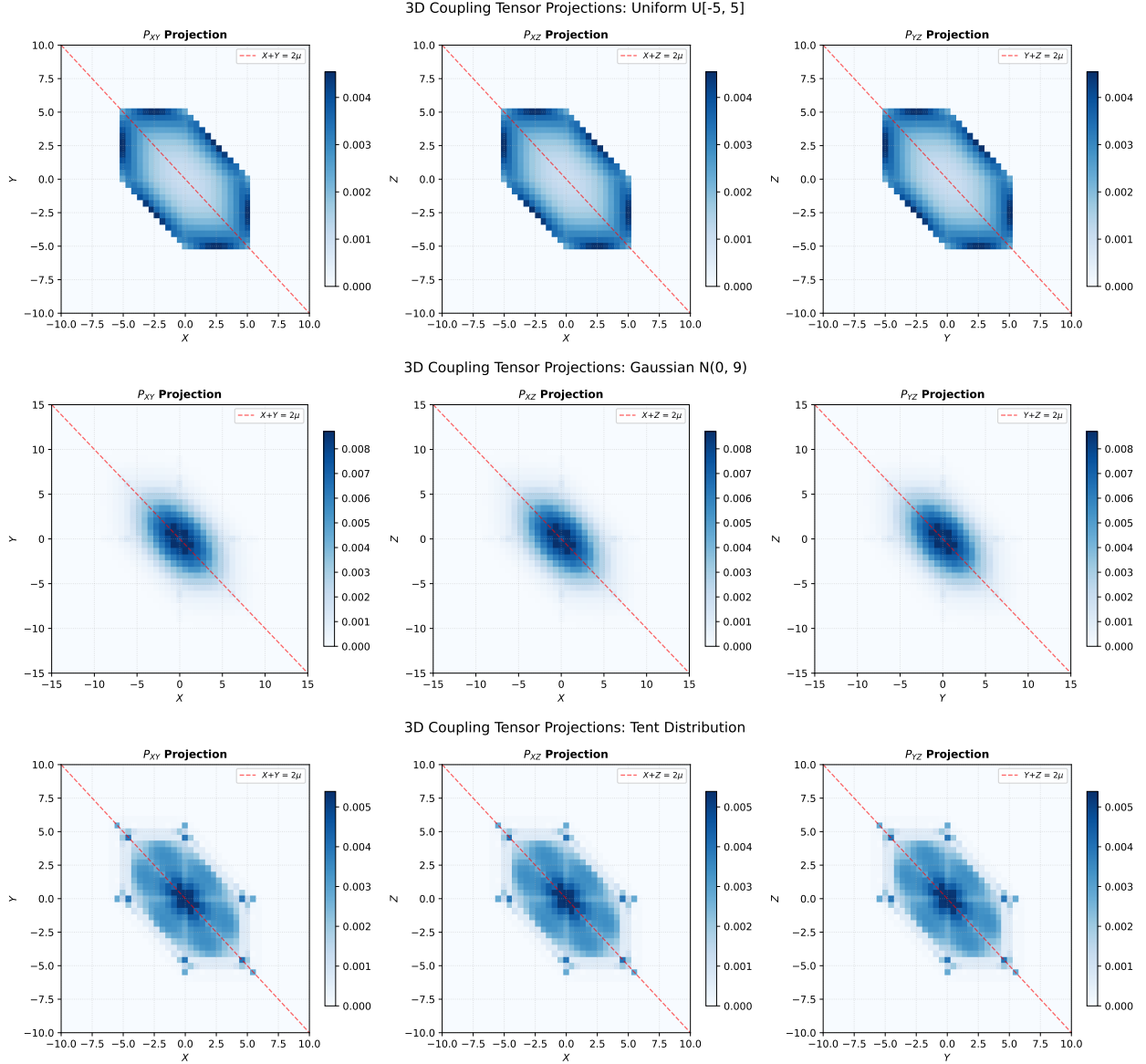


Figure 8: Pairwise projections (P_{XY}, P_{XZ}, P_{YZ}) of the optimal 3-marginal coupling tensor for Uniform $[-5, 5]$ (top), $\mu = N(0, 3^2)$ (middle), and a symmetric tent distribution with mean 0 and radius 6 (bottom); grid size $N = 45$.

2.3 Symmetric bimodal Gaussian mixtures

Let $\mu_{m,\sigma}$ be the equal-weight mixture of $N(-m, \sigma^2)$ and $N(m, \sigma^2)$.

Proposition 1 (Scale invariance of 3-CM under the component Sharpe ratio). *Whether $\mu_{m,\sigma}$ is 3-CM depends on (m,σ) only through m/σ .*

Proof. First note that if $X_1 + X_2 + X_3 = 3\mu_X$ a.s. with $X_i \stackrel{d}{=} F$, and $Y_i = \alpha + \beta X_i$ for $\beta \neq 0$, then $Y_1 + Y_2 + Y_3 = 3\alpha + \beta(X_1 + X_2 + X_3) = 3\alpha + 3\beta\mu_X = 3\mu_Y$ a.s., and $Y_i \stackrel{d}{=} \alpha + \beta F$. We can also apply the same argument in reverse with $(Y_i - \alpha)/\beta$. Hence, 3-CM is invariant under affine transformations. Let $Z \sim N(0,1)$ and S (independent of Z) take values -1 and 1 , each with probability $1/2$. Then $X \sim \mu_{m,\sigma}$ can be represented as $X = S \cdot m + \sigma Z$, so $Y = X/\sigma = S \cdot (m/\sigma) + Z \sim \mu_{m/\sigma,1}$, and $Y = \alpha + \beta X$ with $\alpha = 0, \beta = 1/\sigma \neq 0$. Hence, $\mu_{m,\sigma}$ is 3-CM iff $\mu_{m/\sigma,1}$ is. $\square$

Numerically, we find a 3-CM upper bound of $m/\sigma \approx 2.13$. Below this threshold, the bimodal law is 3-CM and above this threshold it is not. This is illustrated in Figure 9.

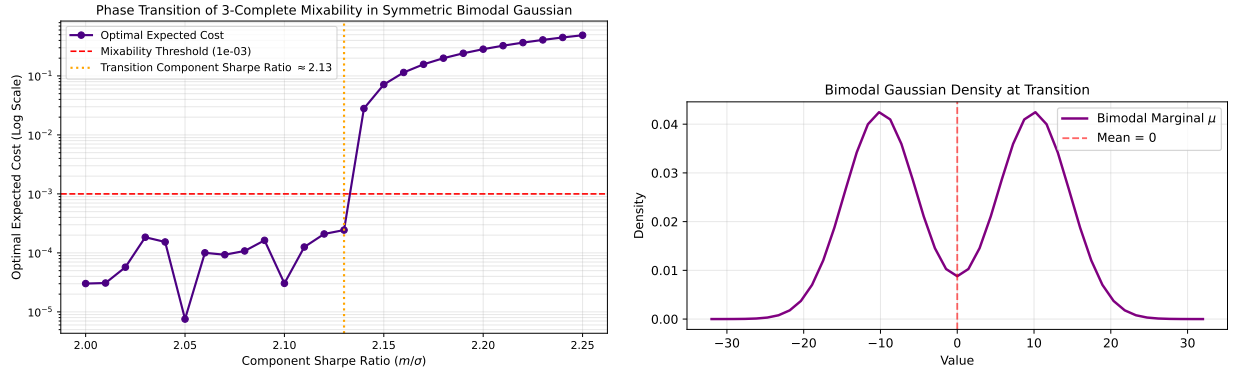


Figure 9: Left: optimal cost against $m/\sigma \in [2.00, 2.25]$ with increments of 0.1, $N = 45$. Right: the symmetric bimodal density at the approximate mixability threshold $m/\sigma \approx 2.13$.

2.4 Symmetric Beta(α, α)

In the case of a symmetric Beta distribution, we find a threshold numerically at $\alpha \approx 0.639$, above which the Beta law is mixable and below which it is not. This is illustrated in Figure 10.

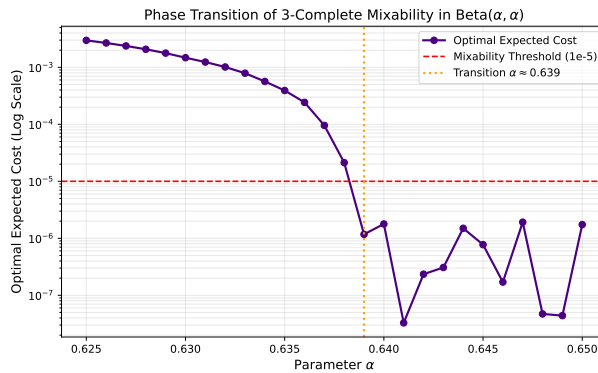


Figure 10: Optimal cost for symmetric Beta(α, α), $\alpha \in [0.625, 0.65]$, $N = 45$. The transition occurs between $\alpha = 0.638$ and $\alpha = 0.639$.

2.5 Symmetric 3-point laws

Let μ_p place mass p at ± 1 and $1 - 2p$ at 0 , $p \in [0, \frac{1}{2}]$. Numerically, we find a 3-CM threshold at $p = 1/3$. That is, the law is 3-CM for all $p \leq 1/3$ but not for $p > 1/3$. We show the projections of the optimal coupling tensor at $p = 0.33$ in Figure 11.

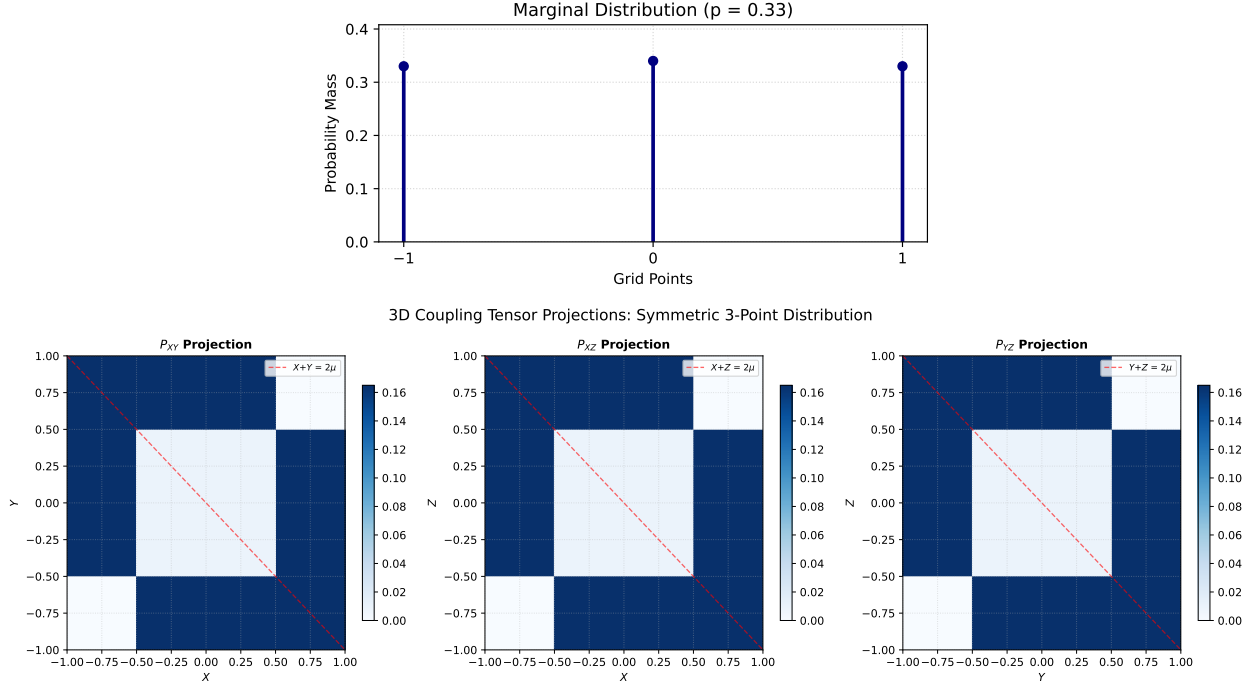


Figure 11: Symmetric 3-point marginal at $p = 0.33$ (top). Corresponding pairwise projections of the 3-point optimal tensor for the law defined by $p = 0.33$ (bottom).

2.6 Symmetric 5-point laws

Let μ_{p_1, p_2} place mass p_2 at ± 2 , p_1 at ± 1 , and $1 - 2p_1 - 2p_2$ at 0 . For example, the marginal defined by $p_1 = 0.07, p_2 = 0.3$, as well as the pairwise projections of the optimal tensor, is illustrated in Figure 12.

Checking 3-CM numerically across a discretised grid of (p_1, p_2) values. We obtain Figure 13 and driven by the patterns observed here, we conjecture the following.

Conjecture 1. *Let μ_{p_1, p_2} place mass p_2 at ± 2 , p_1 at ± 1 , and $1 - 2p_1 - 2p_2$ at 0 ($p_1, p_2 \geq 0$, $p_1 + p_2 \leq 0.5$). Then μ_{p_1, p_2} is 3-CM if and only if $p_1 \leq \frac{1}{3}$ and $p_1 + 2p_2 \leq \frac{2}{3}$.*

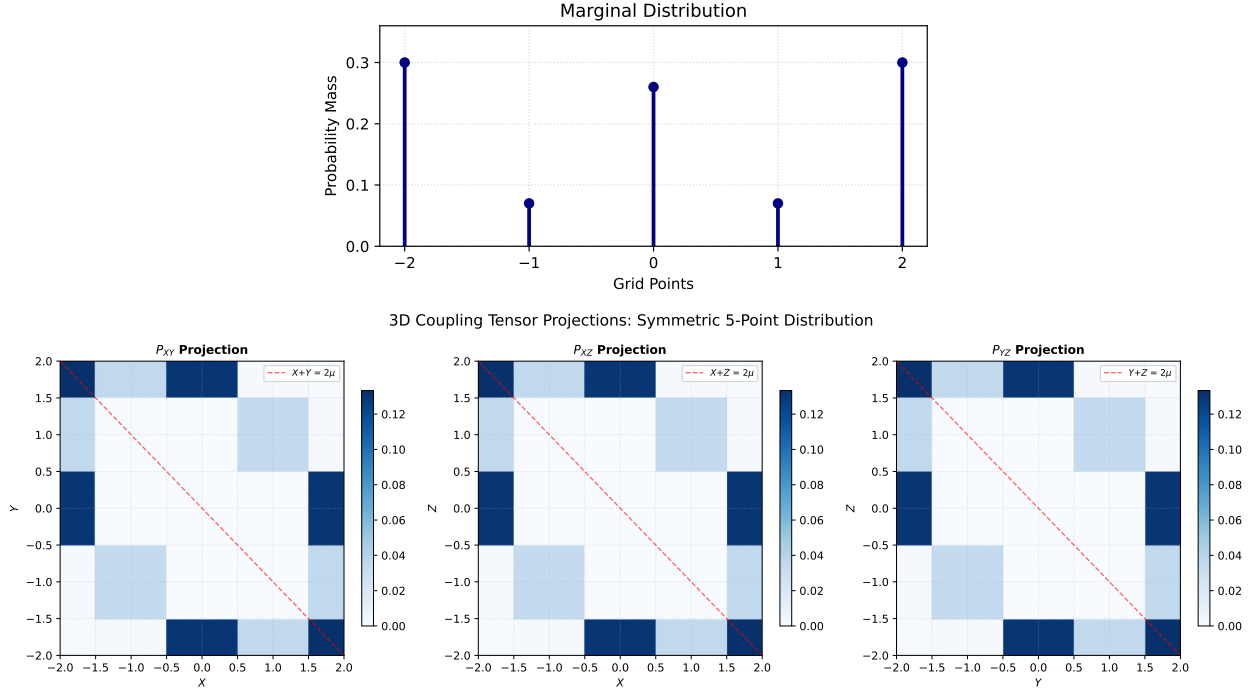


Figure 12: Symmetric 5-point marginal at $p_1 = 0.07, p_2 = 0.3$ (top). Corresponding pairwise projections of the 5-point optimal tensor for the law defined by $p_1 = 0.07, p_2 = 0.3$ (bottom).

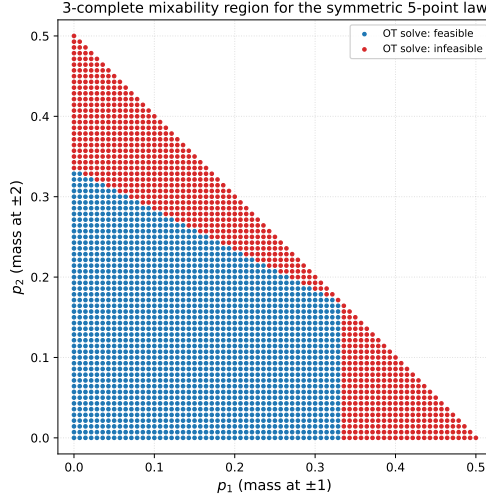


Figure 13: Numerically determined 3-CM region for μ_{p_1, p_2} .

A Numerical Implementation

All problems were solved with `cvxpy` [4] using the CLARABEL interior-point solver [3], tolerances at `tol_gap_abs=tol_gap_rel=tol_feas=1e-7`. Grid sizes: $N = 150$ (two-marginal Asian experiments, Section 1), $N = 45$ (three-marginal Gaussian/uniform/tent/bimodal/Beta experiments), $N = 3$ and $N = 5$ (point-mass experiments). The two core LP templates are given below, the former used for all numerics and figures in Section 1 and the latter for all numerics and figures in Section 2.

```
1 # Two-marginal martingale OT (Section 1)
2 P = cp.Variable((N, N), nonneg=True)
3 constraints = [cp.sum(P, axis=1) == mu,
4               cp.sum(P, axis=0) == nu,
5               P @ grid_y == cp.multiply(mu, grid_x)]
6 objective = cp.Maximize(cp.sum(cp.multiply(P, C)))          # C = |x+y-K|
7               + eps*|x-y|
8 cp.Problem(objective, constraints).solve(solver=cp.CLARABEL, ...)
```

```
1 # Three-marginal (no martingale constraint) OT (Section 2)
2 P = cp.Variable((N, N, N), nonneg=True)
3 constraints = [cp.sum(P, axis=(1, 2)) == mu,
4               cp.sum(P, axis=(0, 2)) == mu,
5               cp.sum(P, axis=(0, 1)) == mu]
6 objective = cp.Minimize(cp.sum(cp.multiply(P, C)))          # C = |x+y+z|
7               - 3*mean|
8 cp.Problem(objective, constraints).solve(solver=cp.CLARABEL, ...)
```

References

- [1] L. Rüschendorf, L. Uckelmann. On the n -coupling problem. *Journal of Multivariate Analysis* 81(2), 242–258, 2002.
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